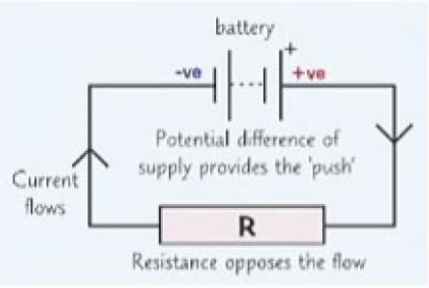
Current and Circuit Symbols

It's pretty bad news if the word current makes you think of delicious cakes instead of physics. Learn what it means, as well as some handy symbols to show items like batteries and switches in a circuit.

Current is the Flow of Electrical Charge



1) Electric current is a flow of electrical charge. Electrical charge will only flow round a complete (closed) circuit if there is a potential difference, so a current can only flow if there's a source of potential difference. The unit of current is the ampere, A.
2) In a single, closed loop (like the one on the right) the current has the same value everywhere in the circuit (see p.28).
3) Potential difference (or voltage) is the driving force that pushes the charge round. Its unit is the volt, V.
4) Resistance is anything that slows the flow down. Unit: ohm, Ω.
5) The current flowing through a component depends on the potential difference across it and the resistance of the component (p.25).



The greater the resistance across a component, the smaller the current that flows (for a given potential difference across the component).

Total Charge Through a Circuit Depends on Current and Time

The size of the current is the rate of flow of charge. When current flows past a point in a circuit for a length of time then the charge that has passed is given by this formula:

Charge flow (coulombs, C) — $Q = It$ — Current (A), Time (s)

More charge passes around the circuit when a larger current flows.

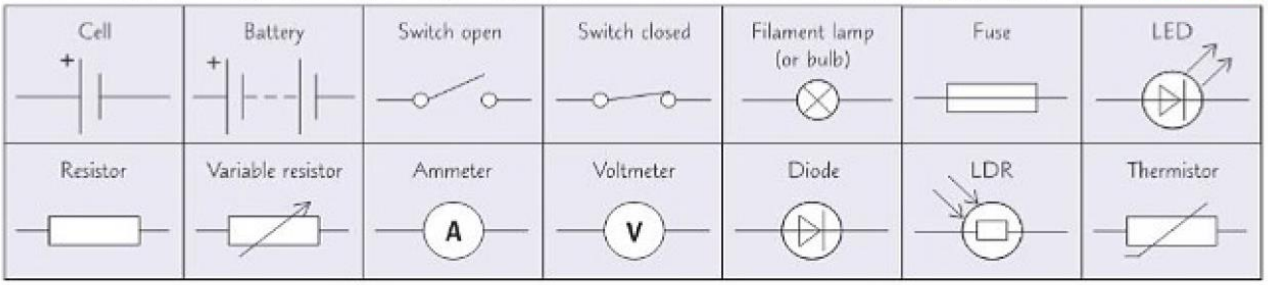
EXAMPLE:

A battery charger passes a current of 2.0 A through a cell over a period of 2.5 hours. How much charge is transferred to the cell?

$Q = It = 2.0 \times (2.5 \times 60 \times 60)$
$= 18\,000$ C

Learn these Circuit Diagram Symbols

You need to be able to understand circuit diagrams and draw them using the correct symbols. Make sure all the wires in your circuit are straight lines and that the circuit is closed, i.e. you can follow a wire from one end of the power supply, through any components, to the other end of the supply (ignoring any switches).



I think it's about time you took charge...

You've no doubt seen some of those circuit symbols before, but take a good look at all of them and practise drawing them. It's no good if you get asked to draw a circuit diagram and you can't tell a resistor from a fuse.

Q1 A laptop charger passes a current of 8.0 A through a laptop battery. Calculate how long the charger needs to be connected to the battery for 28 800 C of charge to be transferred to the laptop. [3 marks]

Q2 A student creates a simple circuit containing a battery, a switch and a bulb. He connects them all in a single, closed loop. Draw the circuit diagram for this circuit. [3 marks]

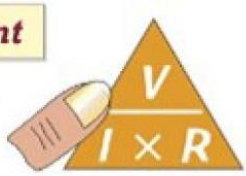
Resistance and V = IR

Ooh experiments, you've gotta love 'em. Here's a simple experiment for investigating resistance.

There's a Formula Linking Potential Difference and Current

The formula linking pd and current is very useful (and pretty common):

Potential difference (V) = Current (A) × Resistance (Ω)



Use this formula triangle to rearrange. Just cover up the thing you're trying to find, and what's left visible is the formula you're after.

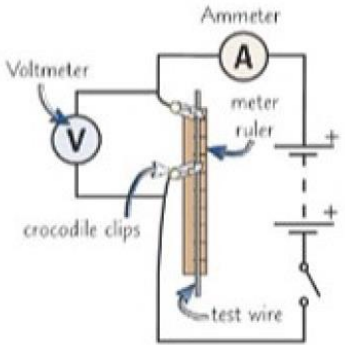
EXAMPLE: A 4.0 Ω resistor in a circuit has a potential difference of 6.0 V across it. What is the current through the resistor?

Rearrange $V = IR$ to give $I = V \div R$, then substitute in the values you have. $I = 6.0 \div 4.0 = 1.5$ A

You Can Investigate the Factors Affecting Resistance

PRACTICAL

The resistance of a circuit can depend on a number of factors, like whether components are in series or parallel, p.31, or the length of wire used in the circuit. You can investigate the effect of wire length using the circuit below.



The Ammeter

1) Measures the current (in amps) flowing through the test wire.
2) The ammeter must always be placed in series with whatever you're investigating.

See p.28-29 for more on series and parallel circuits.

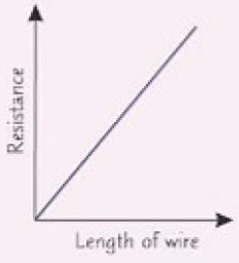
The Voltmeter

1) Measures the potential difference (or pd) across the test wire (in volts).
2) The voltmeter must always be placed in parallel around whatever you're investigating (p.29) — NOT around any other bit of the circuit, e.g. the battery.

A thin wire will give you the best results. Make sure it's as straight as possible so your length measurements are accurate.

1) Attach a crocodile clip to the wire level with 0 cm on the ruler.
2) Attach the second crocodile clip to the wire, e.g. 10 cm away from the first clip. Write down the length of the wire between the clips.
3) Close the switch, then record the current through the wire and the pd across it.
4) Open the switch, then move the second crocodile clip, e.g. another 10 cm, along the wire. Close the switch again, then record the new length, current and pd.
5) Repeat this for a number of different lengths of the test wire.
6) Use your measurements of current and pd to calculate the resistance for each length of wire, using $R = V \div I$ (from $V = IR$).
7) Plot a graph of resistance against wire length and draw a line of best fit.
8) Your graph should be a straight line through the origin, meaning resistance is directly proportional to length — the longer the wire, the greater the resistance.
9) If your graph doesn't go through the origin, it could be because the first clip isn't attached exactly at 0 cm, so all of your length readings are a bit out. This is a systematic error (p.5).

The wire may heat up during the experiment, which will affect its resistance (p.26). Leave the switch open for a bit between readings to let the circuit cool down.



Measure gymnastics — use a vaultmeter...

You could also investigate the effect of diameter or material on the resistance of a wire. What fun.

Q1 An appliance is connected to a 230 V source. Calculate the resistance of the appliance if a current of 5.0 A is flowing through it. [3 marks]

Resistance and I-V Characteristics

Time to investigate just how current varies with potential difference. Then you can make some sweet graphs...

Ohmic Conductors Have a Constant Resistance



For some components, as the current through them is changed, the resistance of the component changes as well.

1) The resistance of ohmic conductors (e.g. a wire or a resistor) doesn't change with the current. At a constant temperature, the current flowing through an ohmic conductor is directly proportional to the potential difference across it. (R is constant in $V = IR$, previous page.)
2) The resistance of some resistors and components DOES change, e.g. a diode or a filament lamp.
3) When an electrical charge flows through a filament lamp, it transfers some energy to the thermal energy store of the filament (p.11), which is designed to heat up. Resistance increases with temperature, so as the current increases, the filament lamp heats up more and the resistance increases.
4) For diodes, the resistance depends on the direction of the current. They will happily let current flow in one direction, but have a very high resistance if it is reversed.

Three Very Important I-V Characteristics

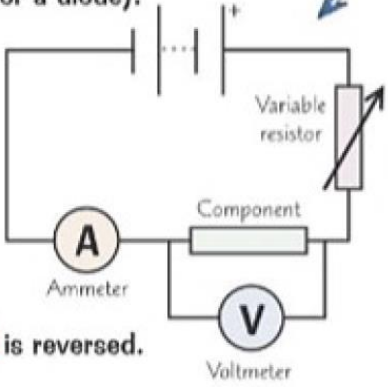
PRACTICAL

The term '*I-V* characteristic' refers to a graph which shows how the current (I) flowing through a component changes as the potential difference (V) across it is increased.
Linear components have an *I-V* characteristic that's a straight line (e.g. a fixed resistor).
Non-linear components have a curved *I-V* characteristic (e.g. a filament lamp or a diode).

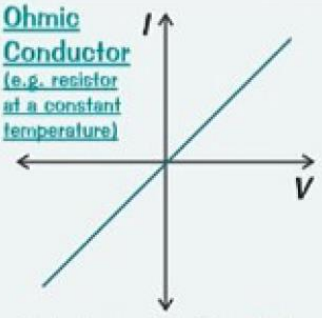
You can do this experiment to find a component's *I-V* characteristic:

1) Set up the test circuit shown on the right.
2) Begin to vary the variable resistor. This alters the current flowing through the circuit and the potential difference across the component.
3) Take several pairs of readings from the ammeter and voltmeter to see how the potential difference across the component varies as the current changes. Repeat each reading twice more to get an average pd at each current.
4) Swap over the wires connected to the cell, so the direction of the current is reversed.
5) Plot a graph of current against voltage for the component.
6) The *I-V* characteristics you get for an ohmic conductor, filament lamp and diode should look like this:

This type of circuit uses direct current (dc) (p.31) and is a series circuit (p.28).

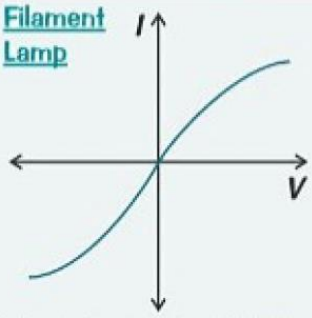


Ohmic Conductor (e.g. resistor at a constant temperature)



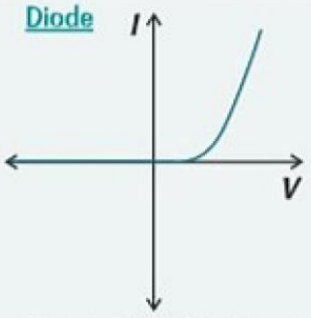
The current through an ohmic conductor (at constant temperature) is directly proportional to potential difference so you get a straight line.

Filament Lamp



As the current increases, the temperature of the filament increases, so the resistance increases. This means less current can flow per unit pd, so the graph gets shallower — hence the curve.

Diode



Current will only flow through a diode in one direction, as shown. The diode has very high resistance in the reverse direction.

Since $V = IR$, you can calculate the resistance at any point on the *I-V* characteristic by calculating $R = V/I$.

In the end you'll have to learn this — resistance is futile...

Draw out those graphs until you're sketching them in your sleep and make sure you can tell whether they're showing a linear or non-linear component.

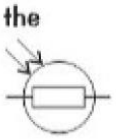
Q1 Draw the *I-V* characteristic for: a) an ohmic conductor b) a filament lamp [4 marks]

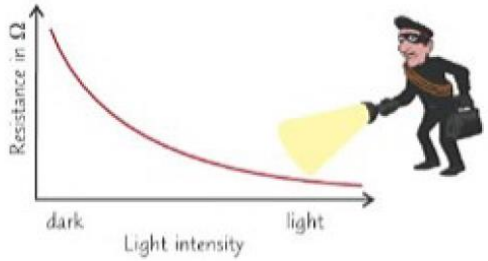
Circuit Devices

The fun doesn't stop with filament bulbs. As well as temperature, resistance can depend on things like light intensity, which is how LDRs work. They're really useful for circuits that sense changes in light levels.

LDR is Short for Light Dependent Resistor

1) An LDR is a resistor that is dependent on the intensity of light. Simple really.
2) In bright light, the resistance falls.
3) In darkness, the resistance is highest.
4) They have lots of applications including automatic night lights, outdoor lighting and burglar detectors.

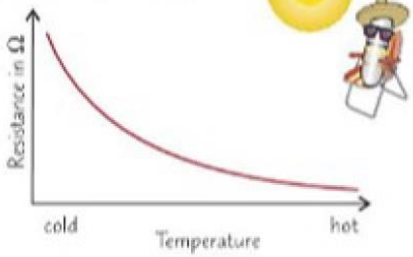




The Resistance of a Thermistor Depends on Temperature

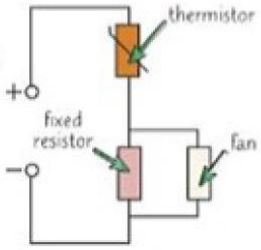
1) A thermistor is a temperature dependent resistor.
2) In hot conditions, the resistance drops.
3) In cool conditions, the resistance goes up.
4) Thermistors make useful temperature detectors, e.g. car engine temperature sensors and electronic thermostats.





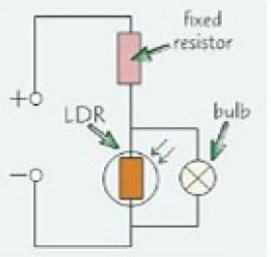
You Can Use LDRs and Thermistors in Sensing Circuits

1) Sensing circuits can be used to turn on or increase the power to components depending on the conditions that they are in.
2) The circuit on the right is a sensing circuit used to control a fan in a room.
3) The fixed resistor and the fan will always have the same potential difference across them (because they're connected in parallel — see p.29).
4) The pd of the power supply is shared out between the thermistor and the loop made up of the fixed resistor and the fan according to their resistances — the bigger a component's resistance, the more of the pd it takes.
5) As the room gets hotter, the resistance of the thermistor decreases and it takes a smaller share of the pd from the power supply. So the pd across the fixed resistor and the fan rises, making the fan go faster.



You can also connect the component across the variable resistor instead.

For example, if you connect a bulb in parallel to an LDR, the pd across both the LDR and the bulb will be high when it's dark and the LDR's resistance is high. The greater the pd across a component, the more energy it gets. So a bulb connected across an LDR would get brighter as the room got darker.



LDRs — Light Dependent Rabbits...

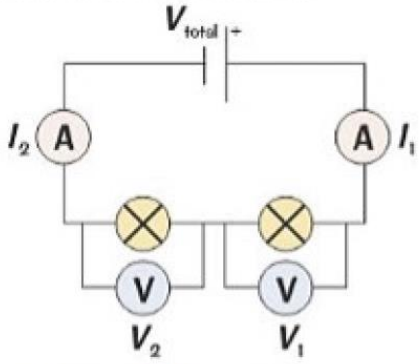
More odd circuit symbols, but at least we're getting into how different components are used in daily life — the next time your heating turns on automatically, you can be smug in your knowledge of thermistors.

Q1 Describe one everyday use for the following components:
a) a LDR b) a thermistor [2 marks]

Series Circuits

There's a difference between connecting components in series and parallel. Make sure you learn it, and know the rules about what happens to current, pd and resistance in each case — read on for more series fun.

Series Circuits — All or Nothing



1) In series circuits, the different components are connected in a line, end to end, between the +ve and –ve of the power supply (except for voltmeters, which are always connected in parallel, but they don't count as part of the circuit).
2) If you remove or disconnect one component, the circuit is broken and they all stop. This is generally not very handy, and in practice very few things are connected in series.
3) You can use the following rules to design series circuits to measure quantities and test components (e.g. the test circuit on p.26 and the sensing circuits on p.27).

Potential Difference is Shared

In series circuits the total pd of the supply is shared between the various components. So the potential differences round a series circuit always add up to equal the source pd:

$$V_{total} = V_1 + V_2 + \ldots$$

Current is the Same Everywhere

1) In series circuits the same current flows through all components, i.e.:

$$I_1 = I_2 = \ldots$$

2) The size of the current is determined by the total pd of the cells and the total resistance of the circuit: i.e. $I = V \div R$.

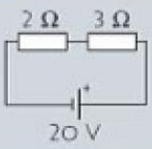
Resistance Adds Up

1) In series circuits the total resistance of two components is just the sum of their resistances:

$$R_{total} = R_1 + R_2$$

2) This is because by adding a resistor in series, the two resistors have to share the total pd.
3) The potential difference across each resistor is lower, so the current through each resistor is also lower. In a series circuit, the current is the same everywhere so the total current in the circuit is reduced when a resistor is added. This means the total resistance of the circuit increases.
4) The bigger a component's resistance, the bigger its share of the total potential difference.

EXAMPLE: For the circuit diagram below, calculate the current passing through the circuit.



1) First find the total resistance by adding together the resistance of the two resistors.
2) Then rearrange $V = IR$ and substitute in the values you have.

$R_{total} = 2 + 3 = 5\ \Omega$
$I = V \div R$
$= 20 \div 5$
$= 4$ A

Cell Potential Differences Add Up

1) There is a bigger pd when more cells are in series, if they're all connected the same way.
2) For example when two cells with a potential difference of 1.5 V are connected in series they supply 3 V between them.

Series circuits — they're no laughing matter...

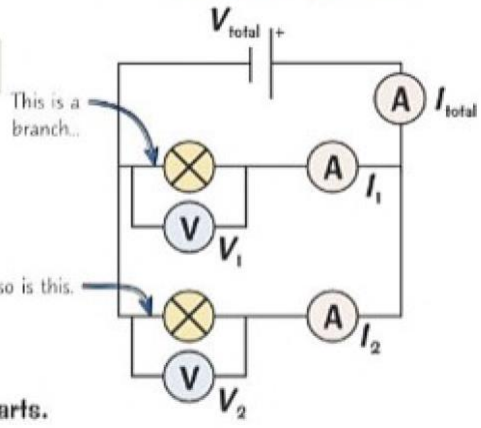
Get those rules straightened out in your head, then have a quick go at this question to test what you can remember.

Q1 Two 12 V cells are connected in series with a 2 Ω resistor, a 3 Ω resistor and a 7 Ω resistor. Calculate the current through the circuit. [5 marks]

Parallel Circuits

Parallel circuits are much more sensible than series circuits and so they're much more common in real life. All the electrics in your house will be wired in parallel circuits.

Parallel Circuits — Independence and Isolation



1) In parallel circuits, each component is separately connected to the +ve and –ve of the supply (except ammeters, which are always connected in series).
2) If you remove or disconnect one of them, it will hardly affect the others at all.
3) This is obviously how most things must be connected, for example in cars and in household electrics. You have to be able to switch everything on and off separately.
4) Everyday circuits often include a mixture of series and parallel parts.

Potential Difference is the Same Across All Components

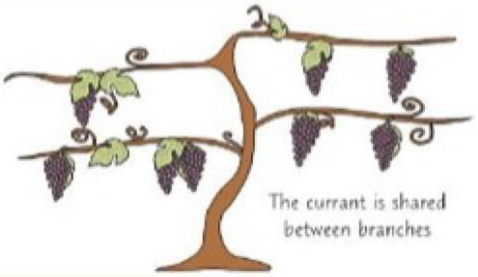
$$V_1 = V_2 = \ldots$$

1) In parallel circuits all components get the full source pd, so the potential difference is the same across all components:
2) This means that identical bulbs connected in parallel will all be at the same brightness.

Current is Shared Between Branches

$$I_{total} = I_1 + I_2 + \ldots$$

1) In parallel circuits the total current flowing around the circuit is equal to the total of all the currents through the separate components.
2) In a parallel circuit, there are junctions where the current either splits or rejoins. The total current going into a junction has to equal the total current leaving.
3) If two identical components are connected in parallel then the same current will flow through each component.



Adding a Resistor in Parallel Reduces the Total Resistance

1) If you have two resistors in parallel, their total resistance is less than the resistance of the smallest of the two resistors.
2) This can be tough to get your head around, but think about it like this:

- In parallel, both resistors have the same potential difference across them as the source.
- This means the 'pushing force' making the current flow is the same as the source pd for each resistor that you add.
- But by adding another loop, the current has more than one direction to go in.
- This increases the total current that can flow around the circuit. Using $V = IR$, an increase in current means a decrease in the total resistance of the circuit.

A current shared *(between identical components)* — is a current halved...

Parallel circuits are a bit more complicated, but they're much more useful than series circuits, so get learning them.

Q1 Explain what happens to the current and resistance in a circuit containing a cell and a resistor when a second resistor is added in parallel. [2 marks]

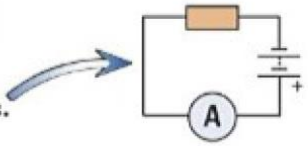
Q2 Draw a circuit diagram for two filament lamps connected in parallel to a battery. Both of the lamps can be switched on and off without affecting each other. [3 marks]

PRACTICAL

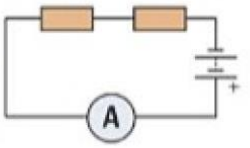
Investigating Resistance

You saw on page 25 how the length of the wire used in a circuit affects its resistance. Now it's time to do an experiment to see how placing resistors in series or in parallel can affect the resistance of the whole circuit.

You Can Investigate Adding Resistors in Series...

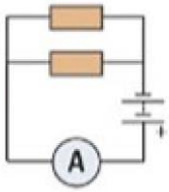


1) First, you'll need to find at least four identical resistors.
2) Then build the circuit shown on the right using one of the resistors. Make a note of the potential difference of the battery (V).
3) Measure the current through the circuit using the ammeter. Use this to calculate the resistance of the circuit using $R = V \div I$.
4) Add another resistor, in series with the first.
5) Again, measure the current through the circuit and use this and the potential difference of the battery to calculate the overall resistance of the circuit.
6) Repeat steps 4 and 5 until you've added all of your resistors.
7) Plot a graph of the number of resistors against the total resistance of the circuit (see below).



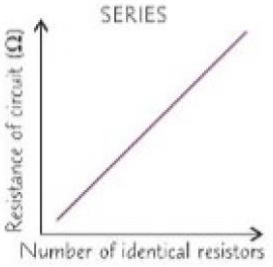
... or in Parallel

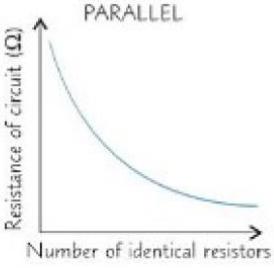
1) Using the same equipment as before (so the experiment is a fair test), build the same initial circuit.
2) Measure the total current through the circuit and calculate the resistance of the circuit using $R = V \div I$ (again, V is the potential difference of the battery).
3) Next, add another resistor, in parallel with the first.
4) Measure the total current through the circuit and use this and the potential difference of the battery to calculate the overall resistance of the circuit.
5) Repeat steps 3 and 4 until you've added all of your resistors.
6) Plot a graph of the number of resistors in the circuit against the total resistance.



Your Results Should Match the Resistance Rules

1) You should find that adding resistors in series increases the total resistance of the circuit (adding a resistor decreases the total current through the circuit).
2) The more resistors you add, the larger the resistance of the whole circuit.
3) When you add resistors in parallel, the total current through the circuit increases — so the total resistance of the circuit has decreased.
4) The more resistors you add, the smaller the overall resistance becomes — as shown by the graph on the right.
5) These results agree with what you learnt about resistance in series and parallel circuits on pages 28 and 29.





I can't resist a good practical...

Nothing too hard on this page, which makes a nice change from all of those rules about circuits. Make sure you're completely happy building circuits from diagrams, before moving on to the fun world of mains electricity... wooo...

Q1 Draw a diagram of a single circuit that could be used to investigate the effect of adding resistors in parallel. Your circuit should include switches. [1 mark]

Electricity in the Home

There are two types of electricity supply — alternating and direct currents. Read on for more about both...

Mains Supply is ac, Battery Supply is dc

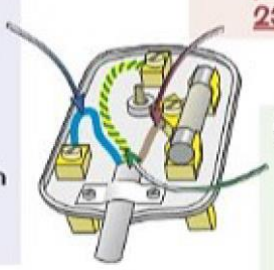
1) There are two types of electricity supplies — alternating current (ac) and direct current (dc).
2) In ac supplies the current is constantly changing direction. Alternating currents are produced by alternating voltages (p.97) in which the positive and negative ends keep alternating.
3) The UK mains supply (the electricity in your home) is an ac supply at around 230 V.
4) The frequency of the ac mains supply is 50 cycles per second or 50 Hz (hertz).
5) By contrast, cells and batteries supply direct current (dc).
6) Direct current is a current that is always flowing in the same direction. It's created by a direct voltage.

Most Cables Have Three Separate Wires

1) Most electrical appliances are connected to the mains supply by three-core cables. This means that they have three wires inside them, each with a core of copper and a coloured plastic coating.
2) The colour of the insulation on each cable shows its purpose.
3) The colours are always the same for every appliance. This is so that it is easy to tell the different wires apart.
4) You need to know the colour of each wire, what each of them is for and what their pd is:

1) LIVE WIRE — brown. The live wire provides the alternating potential difference (at about 230 V) from the mains supply

2) NEUTRAL WIRE — blue. The neutral wire completes the circuit and carries away current — electricity normally flows in through the live wire and out through the neutral wire. It is around 0 V.



3) EARTH WIRE — green and yellow. It is for protecting the wiring, and for safety — it stops the appliance casing from becoming live. It doesn't usually carry a current — only when there's a fault. It's also at 0 V.

The Live Wire Can Give You an Electric Shock

1) Your body (just like the earth) is at 0 V. This means that if you touch the live wire, a large potential difference is produced across your body and a current flows through you.
2) This causes a large electric shock which could injure or even kill you.
3) Even if a plug socket or a light switch is turned off (i.e. the switch is open) there is still a danger of an electric shock. A current isn't flowing but there's still a pd in the live wire. If you made contact with the live wire, your body would provide a link between the supply and the earth, so a current would flow through you.
4) Any connection between live and earth can be dangerous. If the link creates a low resistance path to earth, a huge current will flow, which could result in a fire.



Why are earth wires green and yellow — when mud is brown..?

Electricity is very useful, but it can also be very dangerous. Make sure you know the risks.

Q1 State the potential difference of: a) the live wire b) the neutral wire c) the earth wire. [3 marks]

Power of Electrical Appliances

Energy is transferred between stores electrically (like you saw on page 11) by electrical appliances.

Energy is Transferred from Cells and Other Sources

1) You know from page 11 that a moving charge transfers energy. This is because the charge does work against the resistance of the circuit. (Work done is the same as energy transferred, p.53.)
2) Electrical appliances are designed to transfer energy to components in the circuit when a current flows.

Kettles transfer energy electrically from the mains ac supply to the thermal energy store of the heating element inside the kettle.



Energy is transferred electrically from the battery of a handheld fan to the kinetic energy store of the fan's motor.



3) Of course, no appliance transfers all energy completely usefully. The higher the current, the more energy is transferred to the thermal energy stores of the components (and then the surroundings). You can calculate the efficiency of any electrical appliance — see p.17.

Energy Transferred Depends on the Power

1) The total energy transferred by an appliance depends on how long the appliance is on for and its power.
2) The power of an appliance is the energy that it transfers per second. So the more energy it transfers in a given time, the higher its power.
3) The amount of energy transferred by electrical work is given by:

This equation should be familiar from page 14.

Energy transferred (J) = Power (W) × Time (s) $\quad E = Pt$

EXAMPLE: A 600 W microwave is used for 5 minutes. How long (in minutes) would a 750 W microwave take to do the same amount of work?

1) Calculate the energy transferred by the 600 W microwave in five minutes. $E = Pt = 600 \times (5 \times 60) = 180\,000$ J
2) Rearrange $E = Pt$ and sub in the energy you calculated and the power of the 750 W microwave. $t = E \div P = 180\,000 \div 750 = 240$ s
3) Convert the time back to minutes. $240 \div 60 = 4$ minutes

Remember that the time must be in seconds.

So the 750 W microwave would take 4 minutes to do the same amount of work.

4) Appliances are often given a power rating — they're labelled with the maximum safe power that they can operate at. You can usually take this to be their maximum operating power.
5) The power rating tells you the maximum amount of energy transferred between stores per second when the appliance is in use.
6) This helps customers choose between models — the lower the power rating, the less electricity an appliance uses in a given time and so the cheaper it is to run.
7) But, a higher power doesn't necessarily mean that it transfers more energy usefully. An appliance may be more powerful than another, but less efficient, meaning that it might still only transfer the same amount of energy (or even less) to useful stores (see p.14).

Transfer this page to your useful knowledge store...

Get that equation for power hard-wired into your brain and then become a powerful physicist by practising it:

Q1 An appliance transfers 6000 J of energy in 30 seconds. Calculate its power. [2 marks]

Q2 Calculate the difference in the amount of energy transferred by a 250 W TV and a 375 W TV when they are both used for two hours. [4 marks]

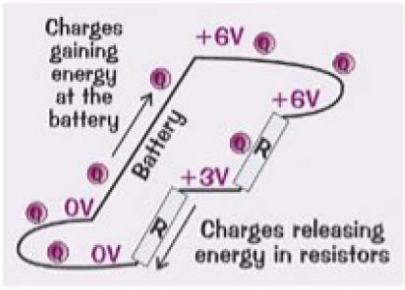
More on Power

And we're not done yet. There are even more power equations for you to get your head around, how fun.

Potential Difference is Energy Transferred per Charge Passed

1) When an electrical charge goes through a change in potential difference, then energy is transferred.
2) Energy is supplied to the charge at the power source to 'raise' it through a potential.
3) The charge gives up this energy when it 'falls' through any potential drop in components elsewhere in the circuit.
4) The formula is real simple:

Energy transferred (J) — $E = QV$ — Charge flow (C), Potential difference (V)



5) That means that a battery with a bigger pd will supply more energy to the circuit for every coulomb of charge which flows round it, because the charge is raised up "higher" at the start.

EXAMPLE:

The motor in an electric toothbrush is attached to a 3 V battery. 140 C of charge passes through the circuit as it is used. Calculate the energy transferred.

$E = QV = 140 \times 3 = 420$ J



This energy is transferred to the kinetic energy store of the motor, as well as to the thermal energy stores of the surroundings.

Power Also Depends on Current and Potential Difference

1) As well as energy transferred in a given time, the power of an appliance can be found with:

Power (W) = Potential difference (V) × Current (A) $\quad P = VI$

EXAMPLE:

A 1.0 kW hair dryer is connected to a 230 V supply. Calculate the current through the hair dryer. Give your answer to two significant figures.

1) Rearrange the equation for current. $I = P \div V$
2) Make sure your units are correct. 1.0 kW = 1000 W
3) Then just stick in the numbers that you have. $I = 1000 \div 230 = 4.34... = 4.3$ A (to 2 s.f.)

2) You can also find the power if you don't know the potential difference. To do this, stick $V = IR$ from page 25 into $P = VI$, which gives you:

$P = I^2R$ — Resistance (Ω)

You have the power — now use your potential...

I'm afraid the best way to learn all of this is to just practise using those equations again and again. Sorry.

Q1 Calculate the energy transferred from a 200 V source as 10 000 C of charge passes. [2 marks]

Q2 An appliance is connected to a 12 V source. A current of 4.0 A flows through it. Calculate the power of the appliance. [2 marks]

Q3 An appliance is connected to a 230 V mains supply and has a current of 10.0 A flowing through it. Calculate the resistance of the appliance. [5 marks]

The National Grid

The national grid is a giant web of wires that covers the whole of Britain, getting electricity from power stations to homes everywhere. Whoever you pay for your electricity, it's the national grid that gets it to you.

Electricity is Distributed via the National Grid

1) The national grid is a giant system of cables and transformers (p.98) that covers the UK and connects power stations to consumers (anyone who is using electricity).
2) The national grid transfers electrical power from power stations anywhere on the grid (the supply) to anywhere else on the grid where it's needed (the demand) — e.g. homes and industry.

Electricity Production has to Meet Demand

1) Throughout the day, electricity usage (the demand) changes. Power stations have to produce enough electricity for everyone to have it when they need it.
2) They can predict when the most electricity will be used though. Demand increases when people get up in the morning, come home from school or work and when it starts to get dark or cold outside. Popular events like a sporting final being shown on TV could also cause a peak in demand.
3) Power stations often run at well below their maximum power output, so there's spare capacity to cope with a high demand, even if there's an unexpected shut-down of another station.
4) Lots of smaller power stations that can start up quickly are also kept in standby just in case.

The National Grid Uses a High Pd and a Low Current

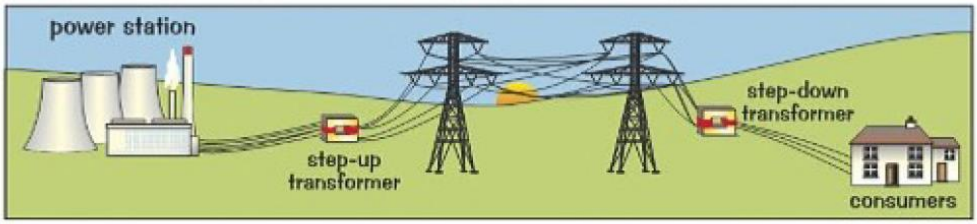
Remember that power is the energy transferred in a given time, so a higher power means more energy transferred.

1) To transmit the huge amount of power needed, you need either a high potential difference or a high current (as $P = VI$, from the previous page).
2) The problem with a high current is that you lose loads of energy as the wires heat up and energy is transferred to the thermal energy store of the surroundings.
3) It's much cheaper to boost the pd up really high (400 000 V) and keep the current as low as possible.
4) For a given power, increasing the pd decreases the current, which decreases the energy lost by heating the wires and the surroundings. This makes the national grid an efficient way of transferring energy.

Potential Difference is Changed by a Transformer

For more on how transformers work, including a useful equation, have a look at page 98.

1) To get the potential difference to 400 000 V to transmit power requires transformers as well as big pylons with huge insulators — but it's still cheaper.
2) The transformers have to step the potential difference up at one end, for efficient transmission, and then bring it back down to safe, usable levels at the other end.



3) The potential difference is increased ('stepped up') using a step-up transformer.
4) It's then reduced again ('stepped down') for domestic use using a step-down transformer.

Transformers — NOT robots in disguise...

Think of this as 'transformers, part one'. You'll run into them again on page 98, but what's important here is that you understand how they're used in the national grid to reduce energy losses during transmission.

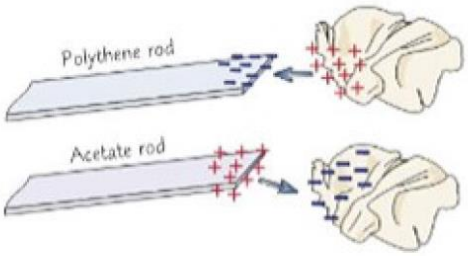
Q1 Explain why the national grid is efficient at transferring energy.
Refer to the potential difference and current during transmission. [4 marks]

Static Electricity

Static electricity is all about charges which are not free to move, e.g. in insulating materials. This causes them to build up in one place and it often ends with a spark or a shock when they do finally move.

Build-up of Static is Caused by Friction

1) When certain insulating materials are rubbed together, negatively charged electrons will be scraped off one and dumped on the other.
2) This will leave the materials electrically charged, with a positive static charge on one and an equal negative static charge on the other.
3) Which way the electrons are transferred depends on the two materials involved.
4) The classic examples are polythene and acetate rods being rubbed with a cloth duster (shown on the right).



Only Electrons Move — Never Positive Charges

Watch out for this in exams. Both +ve and –ve electrostatic charges are only ever produced by the movement of electrons. The positive charges definitely do not move!

A positive static charge is always caused by electrons moving away elsewhere. The material that loses the electrons loses some negative charge, and is left with an equal positive charge. Don't forget!

Too Much Static Causes Sparks

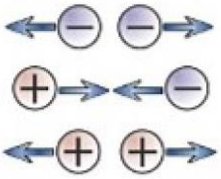
1) As electric charge builds on an object, the potential difference between the object and the earth (which is at 0 V) increases.
2) If the potential difference gets large enough, electrons can jump across the gap between the charged object and the earth — this is the spark.
3) They can also jump to any earthed conductor that is nearby — which is why you can get static shocks getting out of a car. A charge builds up on the car's metal frame, and when you touch the car, the charge travels through you to earth.
4) This usually happens when the gap is fairly small. (But not always — lightning is just a really big spark.)

For more on how sparks actually jump across gaps, see the next page.



Like Charges Repel, Opposite Charges Attract

1) This is easy and, I'd have thought, kind of obvious. When two electrically charged objects are brought close together they exert a force on one another.
2) Two things with opposite electric charges are attracted to each other, while two things with the same electric charge will repel each other.
3) These forces get weaker the further apart the two things are.
4) These forces will cause the objects to move if they are able to do so. This is known as electrostatic attraction / repulsion and is a non-contact force (the objects don't need to touch, p.51).
5) One way to see this force is to suspend a rod with a known charge from a piece of string (so it is free to move). Placing an object with the same charge nearby will repel the rod — the rod will move away from the object. An oppositely charged object will cause the rod to move towards the object.



Stay away from electrons — they're a negative influence...

Electrons jumping about the place and giving us all shocks, the cheeky so-and-sos.

Q1 Jake removes his jumper in a dark room. As he does so, he hears a crackling noise and sees tiny sparks of light between his jumper and his shirt. Explain the cause of this. [3 marks]

Electric Fields

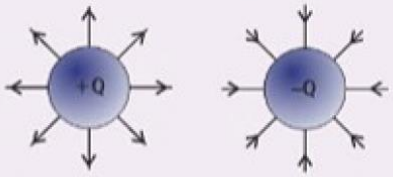
Electric fields — much less green and much more shocking than the fields you're used to.

Electric Charges Create an Electric Field



1) An electric field is created around any electrically charged object.
2) The closer to the object you get, the stronger the field is.
(And the further you are from it, the weaker it is.)
3) You can show an electric field around an object using field lines.
For example you can draw the field lines for an isolated, charged sphere:

Isolated means it's not interacting with anything.



- Electric field lines go from positive to negative.
- They're always at a right angle to the surface.
- The closer together the lines are, the stronger the field is — you can see that the further from a charge you go, the further apart the lines are and so the weaker the field is.

Charged Objects in an Electric Field feel a Force

1) When a charged object is placed in the electric field of another object, it feels a force.
2) This force causes the attraction or repulsion you saw on the previous page.
3) The force is caused by the electric fields of each charged object interacting with each other.
4) The force on an object is linked to the strength of the electric field it is in.
5) As you increase the distance between the charged objects, the strength of the field decreases and the force between them gets smaller.

Two oppositely charged particles



1) The electric field of Q interacts with the electric field of q.
2) This causes forces to act on both Q and q.
3) These forces move q and Q closer together.

Sparking Can Be Explained By Electric Fields



1) Sparks are caused when there is a high enough potential difference between a charged object and the earth (or an earthed object).
2) A high potential difference causes a strong electric field between the charged object and the earthed object.
3) The strong electric field causes electrons in the air particles to be removed (known as ionisation).
4) Air is normally an insulator, but when it is ionised it is much more conductive, so a current can flow through it. This is the spark.

You're one bright spark if you've managed to finish this topic...

And we've saved the toughest 'til last. Re-read this page to really get your head around electric fields. Remember that field lines always point from positive to negative and that the closer they are, the stronger the field is.

Q1 Draw the field lines surrounding an isolated, positively charged sphere. [2 marks]

Q2 Explain what happens to the force acting on a negative charge as it moves closer to a positively charged object. [3 marks]

Revision Questions for Topic 2

Congratulations! You've battled to the end of Topic 2 — now see how much you've learnt.

- Try these questions and tick off each one when you get it right.
- When you've done all the questions under a heading and are completely happy with it, tick it off.

Circuit Basics (p.24-27) ☐

1) Define current and state an equation that links current, charge and time, with units for each. ☐
2) What is meant by potential difference and resistance in a circuit? ☐
3) Draw the circuit symbols for: a cell, a filament lamp, a diode, a fuse and an LDR. ☐
4) What is the equation that links potential difference, current and resistance? ☐
5) Explain how you would investigate how the length of a wire affects its resistance. ☐
6) What is an ohmic conductor? ☐
7) Draw a circuit that could be used to investigate how the resistance of a filament bulb changes with the current through it. ☐
8) Name one linear component and one non-linear component. ☐
9) Explain how the resistance of an LDR varies with light intensity. ☐
10) What happens to the resistance of a thermistor as it gets hotter? ☐

Series and Parallel Circuits (p.28-30) ☐

11) True or false? Potential difference is shared between components in a series circuit. ☐
12) How does the current through each component vary in a series circuit? ☐
13) How does potential difference vary between components connected in parallel? ☐
14) Explain why adding resistors in parallel decreases the total resistance of a circuit, but adding them in series increases the total resistance. ☐
15) Describe an experiment to investigate how adding resistors in series and parallel affects the total resistance of the circuit. ☐

Electricity in the Home (p.31) ☐

16) True or false? Mains supply electricity is an alternating current. ☐
17) What is the potential difference and the frequency of the UK mains supply? ☐
18) Name and give the colours of the three wires in a three-core cable. Why are they colour coded? ☐
19) Give the potential differences for the three wires in a three-core mains cable. ☐
20) Explain why touching a live wire is dangerous. ☐

Power and the National Grid (p.32-34) ☐

21) State three equations that can be used to calculate electrical power. ☐
22) What is the power rating of an appliance? ☐
23) Explain why electricity is transferred by the national grid at a high pd but low current. ☐
24) What are the functions of step-up and step-down transformers? ☐

Static Electricity and Electric Fields (p.35-36) ☐

25) How does rubbing of materials cause static electricity to build up? ☐
26) True or false? Two positive charges attract each other. ☐
27) In which direction do the arrows on electric field lines point? ☐
28) Using the concept of electric fields, explain how a build up of static electricity can cause a spark. ☐